Lesson 5: Induction

ChatGPT edition — motion direction becomes voltage direction

Only changing linked flux produces the pulse

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

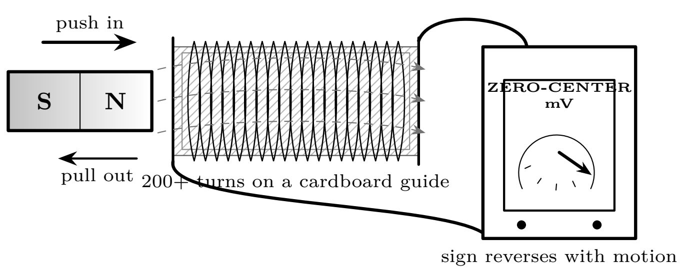
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- Coil of 200+ turns
- Bar magnet
- Sensitive multimeter
- Cardboard guide tube

Predict first

What will the meter show when a magnet rests motionless inside the coil? What changes when motion reverses?



Investigate

1. Connect the unpowered coil to the voltmeter. Hold the magnet still far away, then still inside.
2. Move one pole in at slow and fast repeatable speeds; record sign and peak.
3. Pull it out along the same path. Flip the magnet and repeat.
4. Hold speed approximately fixed and compare one coil with two series coils.

Explain from the evidence

Induced emf follows change in magnetic flux linked through the circuit. No change means no sustained induced signal. Faster change and more linked turns generally increase the response. Reversing motion or pole reverses the sign because the induced effect opposes the change that produces it: Lenz's law and energy conservation tell the same story.

Change one thing

Drop the magnet through a nonconducting guide so speed and path are reproducible; sketch voltage versus time.

Evidence record

Prediction	______
One change	______
Observation	______
Claim + because	______

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.